

1. PROJECT & TEAM INFORMATION

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models
- **Team ID:** SWTID-2026-7649
- **Team Leader:** N.Sopya
- **Team Members:** S S Sathyabala, Mohamed Ajmal, Mohamed Irfan P, Mohamed jubair.P
- **Domain:** Artificial Intelligence & Generative Models

2. INTRODUCTION TO VISUAL STORYTELLING:

Storytelling is one of the oldest and most impactful forms of human communication. Among various storytelling mediums, comic books and graphic novels hold a unique position as they combine the power of textual narrative with compelling visual art. This dual medium allows for a more immersive experience, making complex narratives easier to understand and highly engaging for audiences of all ages. However, the creation of such visual literature has historically been restricted to a small group of professionals who possess a very specific and rare combination of skills: creative writing, character design, storyboarding, and digital or traditional illustration.

3. BACKGROUND OF THE PROBLEM:

In the modern digital era, content creation has been democratized in many fields. Blogging platforms have made anyone a writer, and smartphone cameras have made anyone a photographer. Yet, the creation of comic books remains heavily gated. The traditional pipeline for creating a comic involves multiple laborious stages including scripting, storyboarding, sketching, inking, coloring, and lettering. For an average person with a brilliant story idea, executing this pipeline is nearly impossible without years of artistic training. Hiring freelance illustrators is financially unviable for casual creators, students, and educators.

4. THE CORE PROBLEM STATEMENT:

The central problem this project addresses is the "Visualization Barrier" in creative storytelling. Specifically: "Non-artists, amateur writers, and creative enthusiasts lack accessible, cost-effective, and automated tools to convert their textual story ideas into a structured, visually cohesive, panel-by-panel comic book format."

5. LIMITATIONS OF EXISTING SOLUTIONS :

While Generative AI tools like Midjourney or generic Stable Diffusion interfaces exist, they fail to solve the core problem of comic creation because they lack narrative structure (generating only isolated images), require highly complex prompt engineering, and do not offer automated formatting and layout compilation.

6. PROPOSED SOLUTION: COMICCRAFT AI :

To eradicate these barriers, we propose ComicCraft, a fully automated, end-to-end web application that acts as an AI-powered comic book studio. By integrating Google Gemini AI for structured narrative generation and Hugging Face Stable Diffusion for illustration, ComicCraft allows users to simply input a prompt, character name, and style preference to instantly generate and download a fully formatted, multi-panel comic PDF.